

ETEX

NCSC101

Prof. Pranay Kumar Saha

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What is $\text{\LaTeX}$?

- $\text{\LaTeX}$ is a document preparation system
- Based on $\text{\TeX}$ (created by Donald Knuth in 1978)
- Focuses on content over formatting
- Excellent for:
 - Academic papers
 - Books and technical documentation
 - Mathematical formulas
 - Presentations (like this one!)
- Free and open-source



Basic Document Structure

Minimal L^AT_EX Document

```
\documentclass{article}

\begin{document}
Hello, world!
\end{document}
```

Key Components

- Preamble (before `\begin{document}`)
- Document body (between `\begin{document}` and `\end{document}`)
- Commands start with `\`
- Environments use `\begin{}` and `\end{}`



Document Classes

Common Document Classes

article Scientific journals, presentations, short reports

report Longer reports with chapters, dissertations

book Books

letter Letters

beamer Presentations (like this one)

Syntax

```
\documentclass[options]{class}
```

Common options: **12pt**, **a4paper**, **twocolumn**



Basic Text Formatting

```
Normal text  
\textbf{Bold text}  
\textit{Italic text}  
\underline{Underlined text}  
\texttt{Monospaced text}  
\textsf{Sans serif text}  
\textsc{Small Caps text}  
  
\emph{Emphasized text}
```

Normal text
Bold text
Italic text
Underlined text
Monospaced text
Sans serif text
SMALL CAPS TEXT
Emphasized text



Font Sizes

```
{\tiny Tiny text}  
{\scriptsize Script size}  
{\footnotesize Footnote size}  
{\small Small text}  
{\normalsize Normal size}  
{\large Large text}  
{\Large Larger text}  
{\LARGE Very large text}  
{\huge Huge text}  
  
{\Huge Enormous text}
```

Tiny text

Script size

Footnote size

Small text

Normal size

Large text

Larger text

Very large text

Huge text

Enormous text



Paragraphs and Spacing

First paragraph. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Second paragraph. Line breaks in your source code don't create line breaks in output.

Line break with `\\` creates a new line.
`\\vspace{1cm}` Vertical space
`\\hspace{1cm}` Horizontal space

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Second paragraph. Line breaks in your source code don't create line breaks in output.

Line break with
creates a new line.

Vertical space

Horizontal space after
indentation



Lists

Unordered List

```
\begin{itemize}
  \item First item
  \item Second item
    \begin{itemize}
      \item Nested item
      \item Another nested item
    \end{itemize}
  \item Third item
\end{itemize}
```

Unordered List

- First item
- Second item
 - Nested item
 - Another nested item
- Third item



Lists

Ordered List

```
\begin{enumerate}
  \item First item
  \item Second item
    \begin{enumerate}
      \item Nested item
      \item Another nested item
    \end{enumerate}
  \item Third item
\end{enumerate}
```

Ordered List

1. First item
2. Second item
 - 2.1 Nested item
 - 2.2 Another nested item
3. Third item



Lists

Description List

```
\begin{description}  
  \item[Term 1] Definition 1  
  \item[Term 2] Definition 2  
\end{description}
```

Description List

Term 1 Definition 1
Term 2 Definition 2



Math Mode

Inline Math

Einstein's famous equation
 $E = mc^2$ revolutionized
physics.

The quadratic formula is
 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

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Math Mode

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$$E = mc^2$$

or

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Mathematical Symbols

`$\alpha$, $\beta$, $\gamma$, $\Gamma$`

`$\sum_{i=1}^n i = \frac{n(n+1)}{2}$`

`$\int_a^b f(x) dx$`

`$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$`

`$\nabla \times \vec{F} =`
`$\left( \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i}$`

$$\alpha, \beta, \gamma, \Gamma$$
$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\int_a^b f(x) dx$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\nabla \times \vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} \right) \vec{i}$$



Equation Environments

```
\begin{equation}  
E = mc^2  
\end{equation}
```

$$E = mc^2 \quad (1)$$

```
\begin{align}  
a &= b + c \\\br/>&= d + e \\\br/>&= f  
\end{align}
```

$$a = b + c \quad (2)$$

$$= d + e \quad (3)$$

$$= f \quad (4)$$

```
\begin{cases}  
x + y = 2 \\\br/>x - y = 0  
\end{cases}
```

$$\begin{cases} x + y = 2 \\ x - y = 0 \end{cases}$$



Tables

```
\begin{table}
\caption{Simple Table Example}
\begin{tabular}{|l|c|r|}
\hline
Left & Center & Right \\
\hline
Data & Data & Data \\
More data & More data & More data \\
\hline
\end{tabular}
\end{table}
```

Table: Simple Table Example

Left	Center	Right
Data	Data	Data
More data	More data	More data

- **l** = left-aligned
- **c** = center-aligned
- **r** = right-aligned
- **|** = vertical line



Figures

```
\begin{figure}
  \centering
  \includegraphics[width=0.7\textwidth]
    {image-filename.png}
  \caption{Figure caption
    text}
  \label{fig:example}
\end{figure}
```

Later, refer to Figure
`\ref{fig:example}`.

- Use **graphicx** package
- **\includegraphics** inserts images
- Common image formats: PNG, JPG, PDF
- **\label** and **\ref** for cross-referencing
- Common options:
 - `width=0.8\textwidth`
 - `height=5cm`
 - `scale=0.5`
 - `angle=45`



L^AT_EX

Thank You for Listening!